

6. Transfer of files from one Virtual Machine to another

Aim:

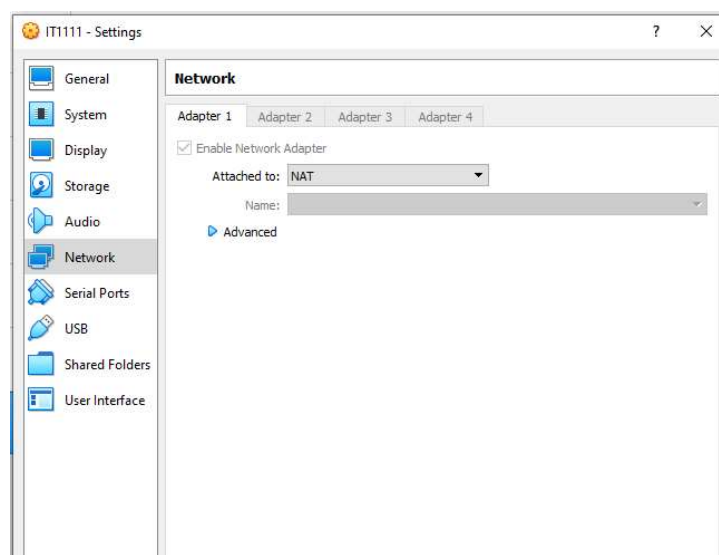
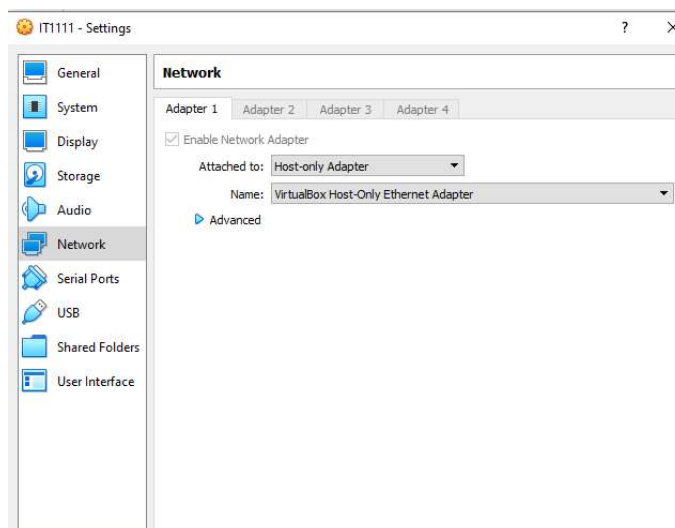
To transfer files from Ubuntu VM to Kali VM using NFS share, enabling read and write access on both ends.

Steps:

1. Configure VirtualBox Network

Set both VMs' network adapter to "Host-only Adapter" in adapter 1 and NAT in adapter 2(for installing packages) so they are on the same subnet.

Go to settings -> network -> adapter 1-> Host only adapter and Adapter2-> NAT



2. Check IP Addresses

On both ubuntu as well as kali ip a (check ip) Ensure both are in same subnet (eg 192.168.56.103/24 ubuntu and 192.168.56.104 kali) both in same subnet 192.168.56.0/24

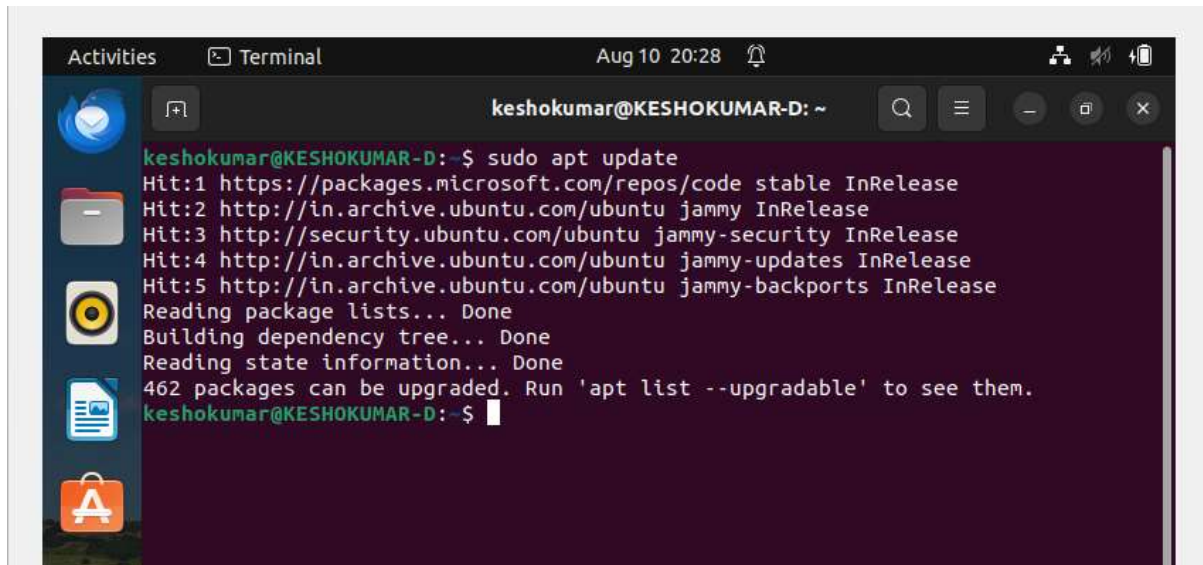
```
tce@tce-VirtualBox:~/praveen111$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:43:b3:7f brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85615sec preferred_lft 85615sec
    inet6 fe80::9637:1bb8:dc16:5da8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 52:54:00:7d:c9:95 brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
        valid_lft forever preferred_lft forever
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel master virbr0 state DOWN group default qlen 1000
    link/ether 52:54:00:7d:c9:95 brd ff:ff:ff:ff:ff:ff
5: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 02:50:96:94:00:00 brd 02:50:96:94:00:00
    inet 172.17.0.1/16 scope global docker0
        valid_lft forever preferred_lft forever
```

```
(tce@TCEDA)-[~/Desktop/praveen111]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:70:32:25 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85807sec preferred_lft 85807sec
    inet6 fe80::a00:27ff:fe70:3225/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(tce@TCEDA)-[~/Desktop/praveen111]
```

3. Set up NFS Server on Ubuntu VM(Here ubuntu act as server)

❖ Sudo apt update



```

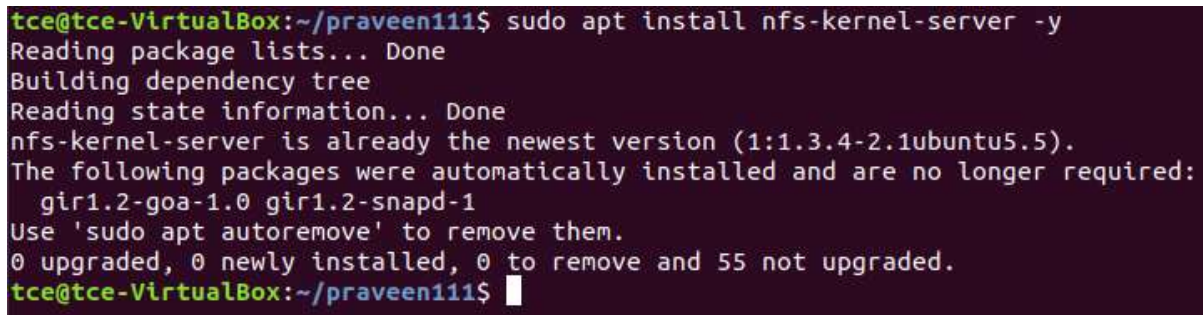
keshokumar@KESHOKUMAR-D: ~
keshokumar@KESHOKUMAR-D:~$ sudo apt update
Hit:1 https://packages.microsoft.com/repos/code stable InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
462 packages can be upgraded. Run 'apt list --upgradable' to see them.
keshokumar@KESHOKUMAR-D:~$

```

❖ Update and install NFS server:

sudo apt update

sudo apt install nfs-kernel-server -y



```

tce@tce-VirtualBox:~/praveen111$ sudo apt install nfs-kernel-server -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
nfs-kernel-server is already the newest version (1:1.3.4-2.1ubuntu5.5).
The following packages were automatically installed and are no longer required:
  gir1.2-goa-1.0 gir1.2-snapd-1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 55 not upgraded.
tce@tce-VirtualBox:~/praveen111$

```

❖ Create folder to share:

sudo mkdir -p /srv/nfs/sharedfolder (or any folder name)

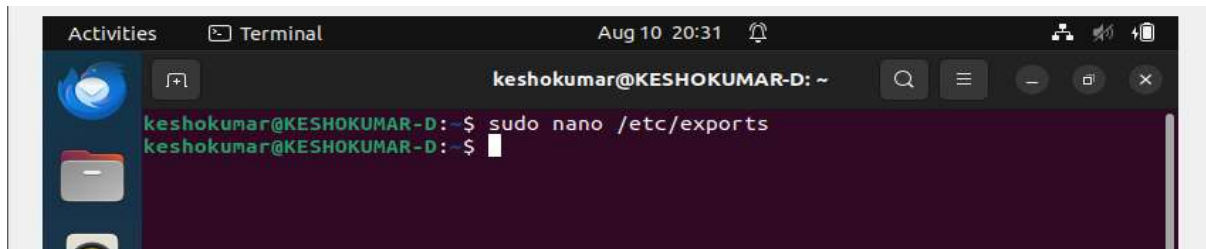
❖ Set permissions:

- sudo chown nobody:nogroup /srv/nfs/sharedfolder
- sudo chmod 777 /srv/nfs/sharedfolder

```
tce@tce-VirtualBox:~/praveen111$ sudo mkdir -p /srv/nfs/sharedfolder
tce@tce-VirtualBox:~/praveen111$ sudo chown nobody:nogroup /srv/nfs/sharedfolder
tce@tce-VirtualBox:~/praveen111$ sudo chmod 777 /srv/nfs/sharedfolder
tce@tce-VirtualBox:~/praveen111$
```

❖ Edit exports file to allow Kali IP access:

`sudo nano /etc/exports` (it open in nano editor)



Add `/srv/nfs/sharedfolder < kali ip >(rw, sync, no_subtree_check, no_root_squash)`

```
GNU nano 2.9.3 /etc/exports Modified
# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients. See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes hostname1(rw, sync, no_subtree_check) hostname2(ro, sync, no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4 gss/krb5i(rw, sync, fsid=0, crossmnt, no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw, sync, no_subtree_check)
#
/srv/nfs/praveen 10.0.2.15(rw, sync, no_subtree_check, no_root_squash)
```

Click `ctrl+o` and enter and `ctrl x` (to save and came out of nano editor)

❖ Export and restart NFS server:

```
sudo exportfs -a
sudo systemctl restart nfs-kernel-server
```

```
tce@tce-VirtualBox:~/praveen111$ sudo exportfs -a
tce@tce-VirtualBox:~/praveen111$ sudo systemctl restart nfs-kernel-server
tce@tce-VirtualBox:~/praveen111$
```

❖ Allow/disable the firewall for the kali linux (skipping shown already firewall in disable mode)

```
tce@tce-VirtualBox:~/praveen111$ sudo ufw allow from 10.0.2.15 to any port nfs
Rules updated
tce@tce-VirtualBox:~/praveen111$
```

4. Set up NFS Client on Kali VM

❖ Install NFS client:

- sudo apt update
- sudo apt install nfs-common -y

(here apt update not work because of network is in off so go to wired connection top right corner change the wired connection(NAT) and after above two command change again the wired connection1(Host only adapter)

```
(tce@TCEDA)-[~/Desktop/praveen111]
$ sudo apt install nfs-common -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-common is already the newest version (1:1.3.4-6).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

(tce@TCEDA)-[~/Desktop/praveen111]
$
```

❖ Create mount point:

```
sudo mkdir -p /mnt/nfs/sharedfolder
```

```
(tce@TCEDA)-[~]
$ sudo mount 192.168.56.108:/srv/nfs/sharedfolder /mnt/nfs/praveen

(tce@TCEDA)-[~]
$
```

❖ Mount the NFS share:

```
sudo mount ubuntu ip:/srv/nfs/dir_name /mnt/nfs/dir_name
```

eg:

```
sudo mount 192.168.56.103:/srv/nfs/sharedfolder /mnt/nfs/sharedfolder
```

```
(tce@TCEDA)-[~]
$ sudo mount 192.168.56.108:/srv/nfs/sharedfolder /mnt/nfs/praveen

(tce@TCEDA)-[~]
$
```

❖ Verify:

```
df -h | grep nfs
```

```
ls -l /mnt/nfs/sharedfolder
```

```
(tce@TCEDA)-[~]
$ df -h | grep nfs
192.168.56.108:/srv/nfs/sharedfolder 49G 16G 32G 33% /mnt/nfs/praveen

(tce@TCEDA)-[~]
$
```

5. Test File Transfer

❖ On Ubuntu VM, create file:

```
echo "Hello from Ubuntu" | sudo tee /srv/nfs/dir_name/ubuntu_test.txt
```

```
tce@tce-VirtualBox:~/praveen111$ echo "Hello from Praveen's Ubuntu" | sudo tee
/srv/nfs/sharedfolder/ubuntu_test.txt
Hello from Praveen's Ubuntu
tce@tce-VirtualBox:~/praveen111$
```

❖ On Kali VM, verify(READ OPERATION)

```
ls -l /mnt/nfs/sharedfolder
```

```
cat /mnt/nfs/sharedfolder/ubuntu_test.txt
```

```
(tce@TCEDA)-[~]
$ cat /mnt/nfs/praveen/ubuntu_test.txt
Hello from Praveen's Ubuntu

(tce@TCEDA)-[~]
$
```

❖ On Kali VM, (WRITE OPERATION):

```
(tce@TCEDA)-[~]
$ echo "Hello from Praveen's Kali" | sudo tee /mnt/nfs/praveen/kali_test.txt
Hello from Praveen's Kali

(tce@TCEDA)-[~]
$
```

❖ On Ubuntu VM, verify

```
ls -l /srv/nfs/sharedfolder
cat /srv/nfs/sharedfolder/kali_test.txt
```

```
tce@tce-VirtualBox:~/praveen111$ cat /srv/nfs/sharedfolder/kali_test.txt
Hello from Praveen's Kali
tce@tce-VirtualBox:~/praveen111$
```

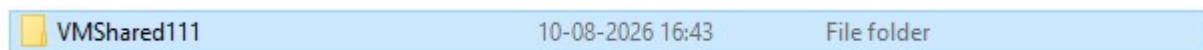
METHOD 2

SHARED FOLDER

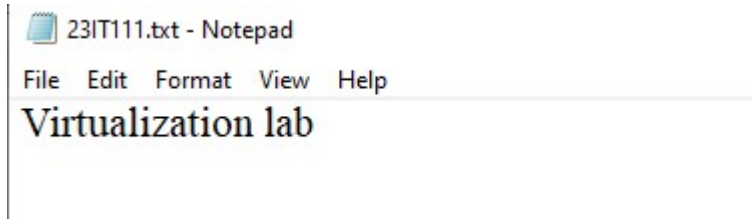
Step 1: Create a Shared Folder on Your Host

1. On your host machine (Windows/Linux):

- Create a folder, e.g., C:\VMShared



- Add a test file (22IT043.txt) to confirm later.



Step 2: Enable Shared Folder in VirtualBox (for Both VMs)

Repeat this for both Ubuntu VMs.

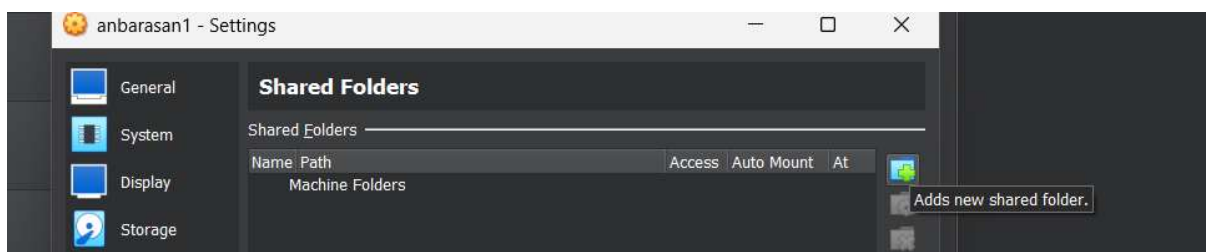
1. Shut down the VM if it's running.
2. In VirtualBox Manager, select your VM.



3. Click Settings → Go to Shared Folders tab.



4. Click “+” (Add new shared folder).



5. Set:

- Folder Path: Select C:\VMShared (or wherever you created it).
- Folder Name: VMShared (updates once path is entered)
- Check(Tick) Auto-mount

Name	Path	Access	Auto Mount	At
Machine Folders				
VMShared111	C:\VMShared111	Full		
Transient Folders				

Click OK.

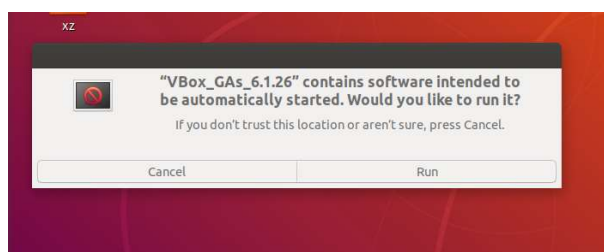
6. Then, again Click OK (at bottom) to save.

Do this for both VMs

VM2:

3.In VirtualBox menu (inside VM window), go to: *Devices* → *Insert Guest Additions CD image*

VM1:



Now use the command: `ls /media`

- It lists all the directories (i.e., mounted devices or drives) under the /media directory.

```
tce@tce-VirtualBox:~$ ls /media
sf_VMShared111  tce
tce@tce-VirtualBox:~$
```

Therefore, in VM1 *sf_VMShared* folder is there in */media* directory.

VM2: SAME PROCEDURE FOR virtualbox 2

Run the command: `ls /media`

```
tce@tce-VirtualBox:~$ ls /media
sf_VMShared111  tce
tce@tce-VirtualBox:~$
```

Therefore, in VM2 also *sf_VMShared* folder is there in */media* directory.

Step 4: Access the Shared Folder

After rebooting each VM, the shared folder should be mounted automatically at:
/media/sf_VMShared

To check, run the command : `ls /media`

VM1 :

```
tce@tce-VirtualBox:~$ ls /media
sf_VMShared111  tce
tce@tce-VirtualBox:~$
```

VM2 :

```
tce@tce-VirtualBox:~$ ls /media
sf_VMShared111  tce
tce@tce-VirtualBox:~$
```

Run the command : `ls /media/sf_VMShared`

VM1 :

```
tce@tce-VirtualBox:~$ ls /media/sf_VMShared111
ls: cannot open directory '/media/sf_VMShared111': Permission denied
tce@tce-VirtualBox:~$
```

VM2 :

```
tce@tce-VirtualBox:~/praveen111$ ls /media/sf_VMShared
ls: cannot access '/media/sf_VMShared': No such file or directory
tce@tce-VirtualBox:~/praveen111$ █
```

You should see the 23IT111.txt file from your host.

But in VM1 and VM2 , it asks for permission to list the files of sf_VMShared folder.

Step 5: Fix Permissions (if Permission Denied)

You need to add your user to the vboxsf group:

```
sudo usermod -aG vboxsf $USER
```

```
tce@tce-VirtualBox:~$ sudo usermod -aG vboxsf $USER
[sudo] password for tce:
tce@tce-VirtualBox:~$ sudo usermod -aG vboxsf $USER
tce@tce-VirtualBox:~$
```

Then reboot again:

```
sudo reboot
```

Now try again:

```
ls /media/sf_VMShared``
```

```
tce@tce-VirtualBox:~$ ls /media/sf_VMShared111
23IT111.txt
█
```

You should now have full access.

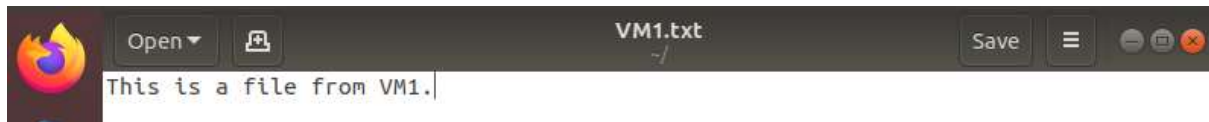
Step 6: Transfer Files Between VMs

- Trying to print the content of 22IT043.txt file from VM1

```
23IT111.txt
tce@tce-VirtualBox:~$ cat /media/sf_VMShared111/23IT111.txt
Virtualization lab
```

- Now in VM1, copy a file to /media/sf_VMShared folder,

```
tce@tce-VirtualBox:~$ touch VM1.txt
tce@tce-VirtualBox:~$ gedit VM1.txt
```



```
cp VM1.txt /media/sf_VMShared/
```

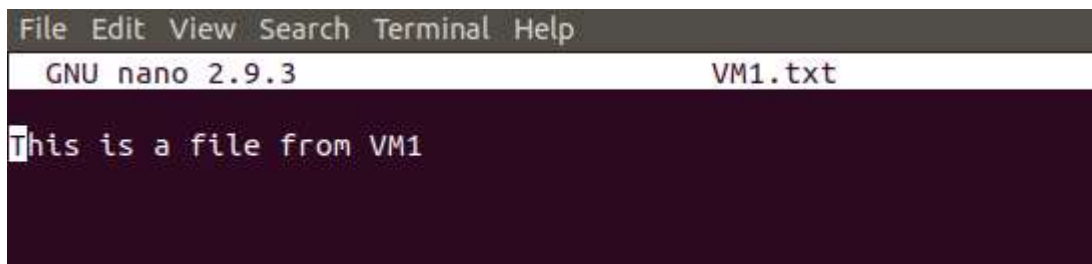
- Switch to VM2, and check the shared folder:

```
ls /media/sf_VMShared
```

The VM1.txt file will appear — you can copy it out to any other location and also you can edit files in shared folder location itself. Any changes done in shared folder using one VM will also be reflected to the other VM.

```
tce@tce-VirtualBox:~$ cp VM1.txt /media/sf_VMShared111/
```

Also 22IT043.txt file is accessible in VM2,



METHOD 3(SSH Secure Shell)

Step 1: Install SSH

On both VMs (VM1 & VM2):

```
sudo apt update
```

```
sudo apt install openssh-server -y
```

```
tce@tce-VirtualBox:~/praveen111$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssh-server is already the newest version (1:7.6p1-4ubuntu0.7).
The following packages were automatically installed and are no longer
gir1.2-goa-1.0 gir1.2-snapd-1
```

Check SSH status:

sudo systemctl status ssh

```
tce@tce-VirtualBox:~/praveen111$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-24 15:34:03 IST; 1min ago
     Main PID: 2953 (sshd)
       Tasks: 1 (limit: 2316)
      CGroup: /system.slice/ssh.service
              └─2953 /usr/sbin/sshd -D

Aug 24 15:34:03 tce-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server: sshd.
Aug 24 15:34:03 tce-VirtualBox sshd[2953]: Server listening on 0.0.0.0 port 22.
Aug 24 15:34:03 tce-VirtualBox sshd[2953]: Server listening on :: port 22.
Aug 24 15:34:03 tce-VirtualBox systemd[1]: Started OpenBSD Secure Shell server: sshd.
```

```
ubuntu@ubuntu1804:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2025-08-22 12:10:18 EDT; 8min ago
     Main PID: 2715 (sshd)
```

Step 2: Get Server's IP

On VM1 (Server, username = tce):*ip a*

```
link/ether 02:42:b9:bb:cb:fb brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
```

Allow firewall settings for SSH to avoid Connection timed out error.

sudo ufw allow 22/tcp

sudo ufw reload

```
tce@tce-VirtualBox:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
tce@tce-VirtualBox:~$ sudo ufw reload
Firewall reloaded
tce@tce-VirtualBox:~$
```

Step 3: Connect from VM2 to VM1

On VM2 (Client, username = ubuntu):

`ssh tce@192.168.56.103`

```
(tce@TCEDA) - [~/Desktop]
$ ssh tce@192.168.56.103
The authenticity of host '192.168.56.108 (192.168.56.108)' can't be established.
ED25519 key fingerprint is: SHA256:CguX5tRRca45dpqts80PS/5y76r8pAHgFxR49A/fbKY
tce@tce-VirtualBox:~$ ls
23IT099  examples.desktop  kavin.java  number.java  Public  VM1.txt
balaji-157  fl.sh  Music  number.java.save  snap
```

Enter tce's password → now you are logged into VM1.

Step 4: Transfer Files with SCP (From VM2)

(a) Copy file from VM2 → VM1

`scp file.txt tce@192.168.56.103:/home/tce/`

```
tce@tce-VirtualBox:~/praveen111$ scp VM2.txt tce@192.168.56.108:/home/tce
tce@192.168.56.108's password:
VM2.txt 100% 18 11.4KB/s 00:00
```

METHOD 4(FTP (File Transfer Protocol))

Step 1: Install FTP Server on VM1

(VM1 = server, VM2 = client)

On VM1:

`sudo apt update`

`sudo apt install vsftpd -y`

```
tce@tce-VirtualBox:~$ sudo apt install vsftpd -y
[sudo] password for tce:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
gir1.2-goa-1.0 gir1.2-spand-1 libmemcached11 libmemcachedutil2 proftpd-doc
```

Enable and start the service:

```
sudo systemctl enable vsftpd
```

```
sudo systemctl start vsftpd
```

```
tce@tce-VirtualBox:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd/sd/terminal: sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable vsftpd
tce@tce-VirtualBox:~$ cd praveen111
tce@tce-VirtualBox:~/praveen111$ sudo systemctl start vsftpd
tce@tce-VirtualBox:~/praveen111$
```

Step 2: Configure vsftpd

Edit the config file:

```
sudo nano /etc/vsftpd.conf
```

```
tce@tce-VirtualBox:~$ sudo nano /etc/vsftpd.conf
[sudo] password for tce:
```

Make sure these lines are set (uncomment or add if missing):

```
local_enable=YES
```

```
write_enable=YES
```

```
chroot_local_user=YES
```

```
allow_writeable_chroot=YES
```

Save (CTRL+S, CTRL+X)

```
tce@tce-VirtualBox:~$ grep -v "^#" /etc/vsftpd.conf
listen=NO
listen_ipv6=YES
anonymous_enable=NO
local_enable=YES
write_enable=YES
dirmessage_enable=YES
use_localtime=YES
xferlog_enable=YES
connect_from_port_20=YES
chroot_local_user=YES
secure_chroot_dir=/var/run/vsftpd/empty
pam_service_name=vsftpd
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
ssl_enable=NO

allow_writeable_chroot=YES
```

Restart the ftp server:

sudo systemctl restart vsftpd

```
tce@tce-VirtualBox:~/praveen111$ sudo nano /etc/vsftpd.conf
tce@tce-VirtualBox:~/praveen111$ sudo systemctl restart vsftpd
```

Step 3: Check VM1 IP Address

On VM1, run:

ip a

```
inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
    valid_lft 364sec preferred_lft 364sec
```

Allow firewall settings for FTP to avoid Connection timed out error.

sudo ufw allow 21/tcp

```
tce@tce-VirtualBox:~/praveen111$ sudo ufw allow 21/tcp
Rule added
Rule added (v6)
tce@tce-VirtualBox:~/praveen111$ sudo ufw allow 20/tcp
Rule added
Rule added (v6)
tce@tce-VirtualBox:~/praveen111$ sudo ufw reload
Firewall reloaded
tce@tce-VirtualBox:~/praveen111$
```

Step 4: Connect from VM2 (FTP client)

On VM2, install ftp client: *sudo apt install ftp -y*

```
ubuntu@ubuntu1804:~$ sudo apt install ftp -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

Now connect:

ftp 192.168.56.103

Enter VM1's username (tce) and password when prompted.

```
(tce@TCEDA)-[~/Desktop]
$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPD 3.0.3)
Name (192.168.56.101:tce): ftpraveen
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
```

Step 5: FTP Commands

Once connected, you can use.

- *cd folder* → change directory on VM1

```
ftp> cd praveen111
250 Directory successfully changed.
ftp> █
```

- *get filename* → download file from VM1 → VM2

```
ftp> ls
227 Entering Passive Mode (192,168,56,101,100,2)
150 Here comes the directory listing.
-rw-r--r--  1 1002    1002           8980 Aug 31 13:44 examples.desktop
-rw-r--r--  1 1002    1002           35 Aug 31 13:45 test.txt
226 Directory send OK.
ftp> get test.txt
local: test.txt remote: test.txt
227 Entering Passive Mode (192,168,56,101,115,48)
150 Opening BINARY mode data connection for test.txt (35 bytes).
226 Transfer complete.
35 bytes received in 0.00 secs (8.2540 kB/s)
ftp> █
```

- *put filename* → upload file from VM2 → VM1

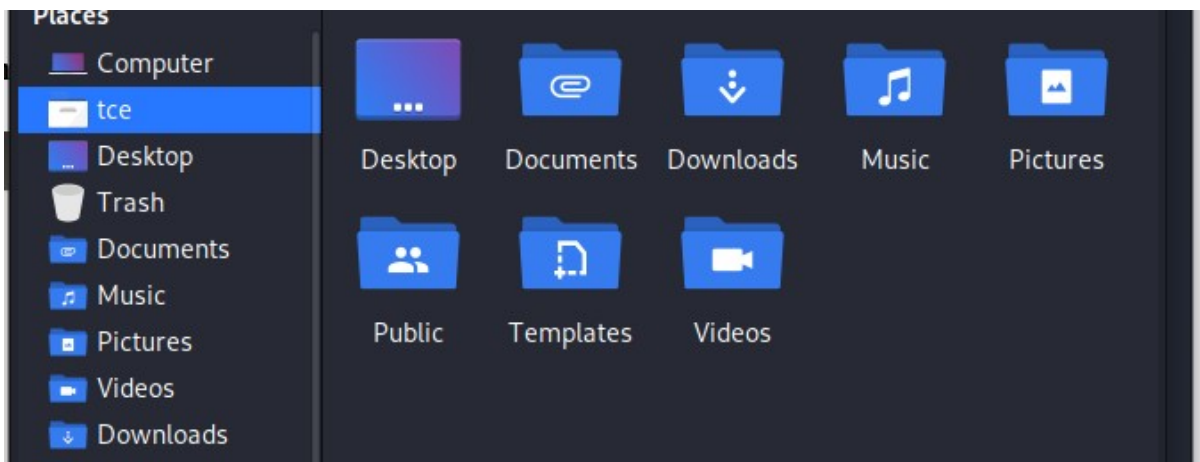
```

local: ftpKali.txt: No such file or directory
ftp> put ftpKali.txt
local: ftpKali.txt remote: ftpKali.txt
227 Entering Passive Mode (192,168,56,101,66,255)
150 Ok to send data.
226 Transfer complete.
ftp> █

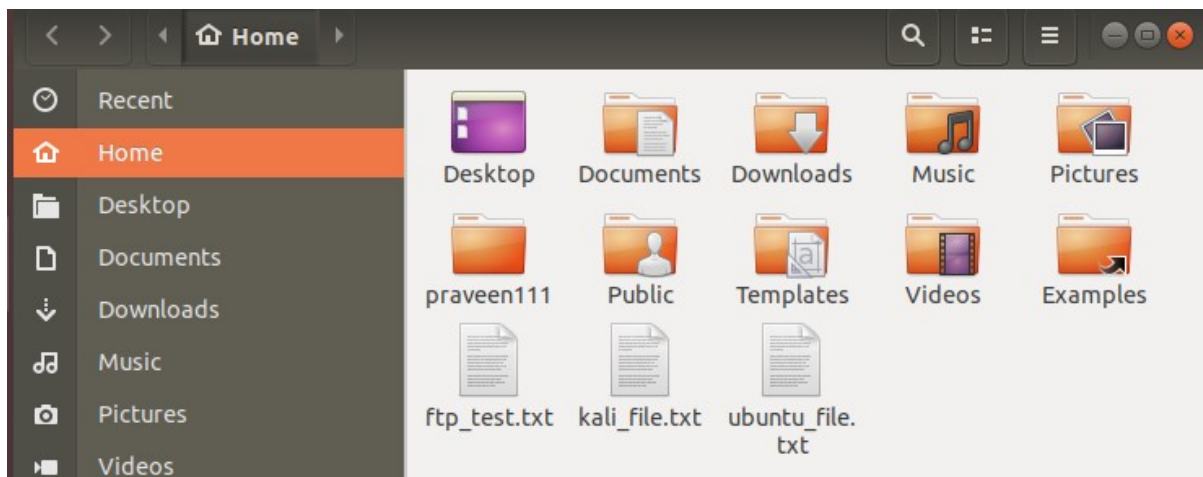
```

Step 6: Verify

- Ubuntu filesystem should now be in VM2's current directory.



- ftpKali.txt should now be uploaded to VM1's */home/FTP1/* directory.



RESULT:

Thus the file transfer between Ubuntu and Kali VMs is executed successfully and both VMs can read and write files in the shared folder seamlessly.